Financial News Sentiment Prediction using Deep Learning & BERT

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Mini Project 5

1. Introduction

Sentiment analysis is crucial in financial markets for understanding investor behavior and market trends. This project focuses on classifying financial tweets into Bearish, Bullish, and Neutral sentiments using deep learning and transformer-based models.

2. Dataset Description

The dataset contains financial tweets labeled into three classes. The dataset is imbalanced, with Neutral class dominating. This posed challenges for traditional models and required careful evaluation.

3. Data Preprocessing

Text preprocessing included cleaning, tokenization, and padding for RNN models. For BERT, pretrained tokenization was used, eliminating manual feature engineering.

4. Models Implemented

LSTM struggled due to class imbalance. GRU improved performance. Bi-GRU captured bidirectional context. BERT significantly outperformed all models due to contextual embeddings and attention mechanism.

5. Model Performance Comparison

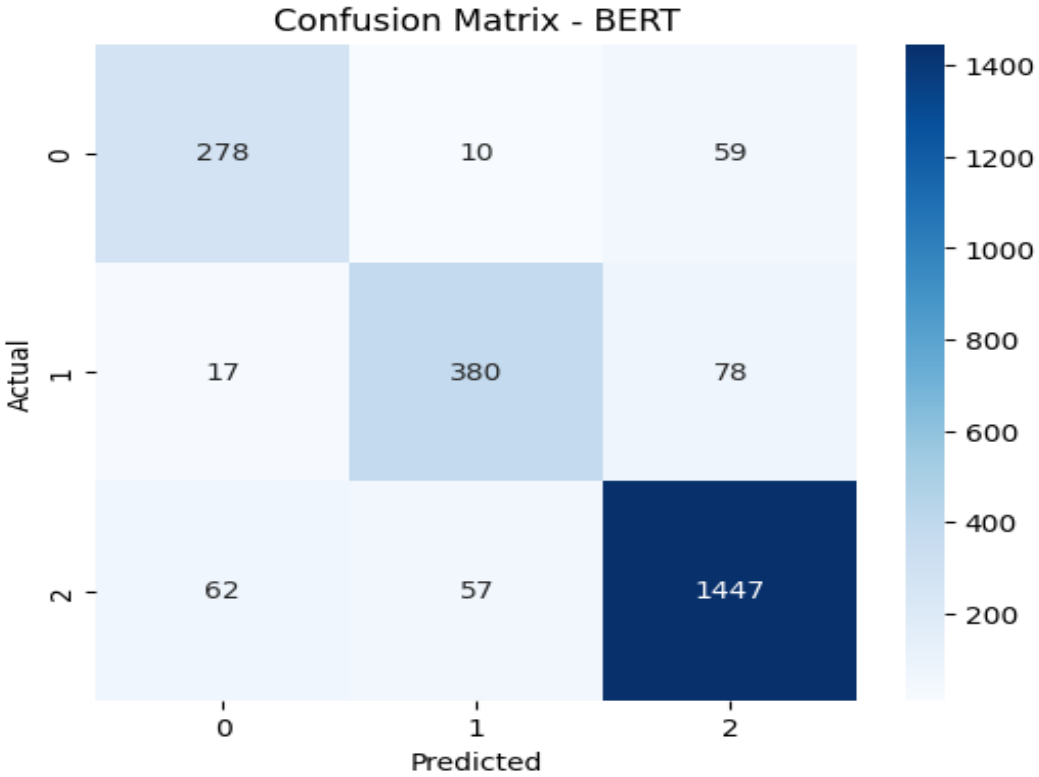
Model	Accuracy	Macro F1
LSTM	66%	0.26
GRU	79%	0.71
Bi-GRU	80%	0.73
BERT	88%	0.85

6. Performance Analysis

Bi-GRU performed best among RNN models due to bidirectional context. BERT improved Macro F1 by ~12% over GRU, showing superior contextual understanding.

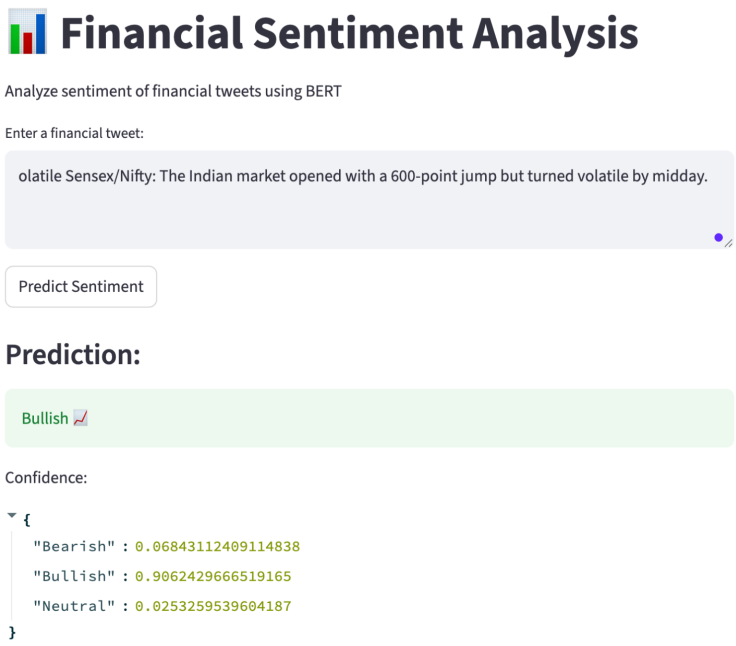
7. Confusion Matrix (BERT)

The confusion matrix demonstrates strong performance across all classes with high correct predictions.



8. Streamlit Application

A Streamlit app was developed to provide real-time sentiment predictions. Users can input text and receive predictions instantly.



9. Conclusion

BERT significantly outperforms RNN models and provides robust sentiment predictions. The project demonstrates a complete end-to-end pipeline from data preprocessing to deployment.

10. Future Improvements

Future work includes FinBERT, hyperparameter tuning, advanced imbalance handling, and cloud deployment.